REDES DE INTERNET (RI)

TRABALHO Nº 2 - OSPF LAB: ADVANCED ROUTING CONFIGURATIONS

INTRODUCTION

Welcome to the Advanced OSPF Configuration Lab. In this lab, you will delve into the intricacies of the Open Shortest Path First (OSPF) routing protocol, one of the most widely used interior gateway protocols in enterprise networks. You'll configure a complex multi-area OSPF network, implement advanced features, and troubleshoot common issues. This hands-on experience will solidify your understanding of OSPF operations and prepare you for real-world network engineering challenges.

LAB OBJECTIVES AND LEARNING OUTCOMES

By the end of this lab, you will be able to:

- Configure OSPF in a multi-area environment
- Implement OSPF authentication to enhance network security
- Set up OSPF virtual links to maintain area connectivity
- Control OSPF route propagation and implement route summarization
- Redistribute routes between OSPF and other routing protocols (specifically RIP)
- Analyse OSPF Link State Advertisements (LSAs) and understand their role in routing decisions
- Troubleshoot common OSPF configuration issues
- Optimize OSPF performance through careful metric and timer adjustments

PREREQUISITES

To successfully complete this lab, you should have:

- Basic understanding of IP routing concepts and subnetting
- Familiarity with Cisco IOS command-line interface
- Basic knowledge of OSPF terminology and operations
- Experience with basic router configuration tasks

LAB ENVIRONMENT

This lab utilizes the following environment:

- Simulation Software: GNS3 (Graphical Network Simulator-3)
- Devices: 10 Cisco routers (series 7200)
- Connections: Gigabit Ethernet interfaces between routers
- Host Machines: PCs connected to specific routers for testing connectivity

Ensure you have access to GNS3 and the required router images before beginning the lab.

NETWORK TOPOLOGY OVERVIEW

The lab network consists of 10 routers organized into 4 OSPF areas:

- Area 0 (Backbone): R1, R2, R3, R4
- Area 1: R7, R5
- Area 2: R2, R8, R9
- Area 3: R5, R6, R1, R3

Key features of the topology:

- Area 1 will require a virtual link to connect to Area 0;
- Area 2 will involve route redistribution with RIP;
- Area 3 connects to the backbone through multiple routers.

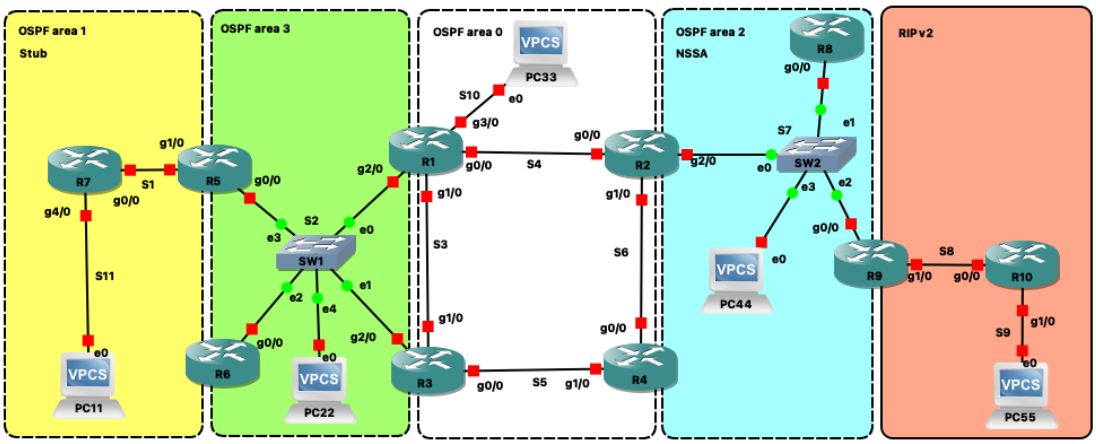


FIGURE 1 - NETWORK TOPOLOGY DIAGRAM

LAB STRUCTURE

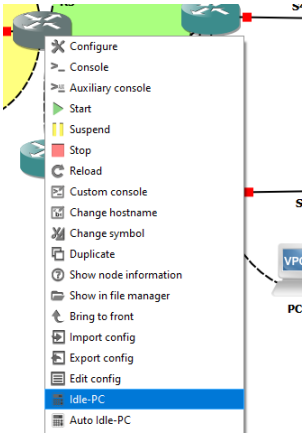
The lab, with the topology visible in Figure 1, is divided into five major sections:

1. Basic OSPF Configuration (Area 0 and Area 3):
   1.1. Configure router IDs, enable OSPF on interfaces, implement authentication;
   1.2. Set up DR/BDR election, configure point-to-point links.
2. OSPF Area 1 and Virtual Link Configuration:
   2.1. Configure Area 1, implement a virtual link to Area 0;
   2.2. Verify inter-area connectivity.
3. OSPF Area 2 and External Route Redistribution:
   3.1. Configure RIP, redistribute RIP routes into OSPF;
   3.2. Control LSA propagation, adjust OSPF metrics for external routes.
4. Advanced OSPF Configurations:
   4.1. Control summary LSAs, optimize route table size;
   4.2. Simulate a maintenance scenario and analyse routing behaviour changes.
5. OSPF optimization and troubleshooting:
   5.1. Apply traffic engineering strategies;
   5.2. Perform failure analysis.

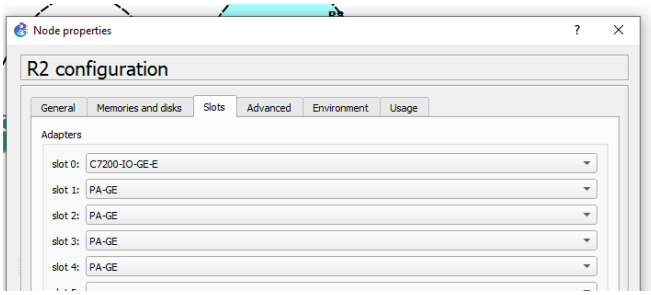
Each section includes specific tasks, verification steps, and review questions to reinforce your learning.

Lab Rules

- Do not use any static routes, default routes, or policy routing unless otherwise specified.
- Save your configuration often `#write mem`.
- Use OSPF process id=1 (router ospf 1) in all the routers.
- The Loopback networks should appear in the routing table with a mask of /32.
- OSPF process should not try to establish a neighbour adjacency through interfaces where adjacencies are not expected (ex: loopbacks).
- Configure the router IDs with the IP address that represent the most stable choice.
- For all the ethernet interfaces with /30 (point to point) choose the OSPF network type that will allow the fastest convergence on neighbour relations establishment.
- In case of high CPU on the GNS3, run the idle PC in the several routers from the topology.



- All the interfaces between the routers on the topology are GE, for this the routers are equipped with the following adapters ex:



- For the network configurations use the following IP addresses:

Loopbacks	IP address
R1	1.1.1.1/32
R2	2.2.2.2/32
R3	3.3.3.3/32
R4	4.4.4.4/32
R5	5.5.5.5/32
R6	6.6.6.6/32
R7	7.7.7.7/32

R8	8.8.8.8/32
R9	9.9.9.9/32
R10	10.10.10.10/32

Segments	Net IP address	Mask
S1	10.7.5.0	/30
S2	10.5.6.0	/29
S3	10.1.3.0	/30
S4	10.1.2.0	/30
S5	10.3.4.0	/30
S6	10.2.4.0	/30
S7	10.2.8.0	/29
S8	10.9.10.0	/30
S9	10.10.55.0	/30
S10	10.1.33.0	/30
S11	10.7.11.0	/30

Expectations and Guidelines

- Complete all tasks in each section, documenting your configurations and observations
- Answer all review questions at the end of each section
- Prepare a lab report including:
  - Your configurations for each task
  - Answers to review questions
  - Analysis of OSPF behaviour observed during the lab
  - Any challenges encountered and how you resolved them
- Submit your lab report at moodle, check the deadlines!
- You may collaborate with classmates on understanding concepts, but all configurations and written work must be from you or your group members.
- At the end of this document is a suggested template for you to use on your report.

Remember, OSPF is a fundamental protocol in many enterprise networks. The skills you develop in this lab will be directly applicable to your future work as a network engineer. Approach each task with curiosity and attention to detail, and don't hesitate to ask questions if you need clarification.

Lab Setup

Software Requirements

Ensure you have the following software installed on your computer:

- GNS3 (version 2.2.0 or later)
- GNS3 VM (if using GNS3 VM – not required)
- Cisco IOS images for the routers (as specified by your teacher)

Importing the Lab Topology:

1. Launch GNS3;
2. Go to File > Import project;
3. Select the provided .gns3 project file for this lab;
4. Wait for the project to load and verify that you see 10 routers in the topology.

Initial Router Configuration

Each router in the topology needs basic configuration. Follow these steps for each router:

1. Right-click on the router and select "Console" to open the CLI
2. Enter privileged EXEC mode: `enable`
3. Enter global configuration mode: `configure terminal`
4. Set the hostname according to the topology (R1, R2, etc.):

```
Router(config)# hostname R1
```

5. Disable DNS lookup to prevent typing mistakes from causing delays:

```
R1(config)# no ip domain-lookup
```

6. Set a password for privileged EXEC mode:

```
R1(config)# enable secret class
```

7. Configure console line:

```
R1(config)# line console 0
R1(config)# password cisco
R1(config)# login
```

8. Configure VTY lines:

```
R1(config)# line vty 0 4
R1(config)# passsword cisco
R1(config)# login
```

9. Save the configuration:

```
R1(config)# end
R1# write memory
```

Lab Environment

1. Since we are running a virtual lab and for being more efficient, we can skip passwords for now and go straight to a privileged environment with the following configuration.

```
no enable secret
line con 0
 exec-timeout 0 0
```

```
 privilege level 15
 no password
```

Interface Configuration

Configure the interfaces on each router according to the IP addressing scheme provided in the topology diagram. Here's an example for R1:

1. Enter global configuration mode: `configure terminal`
2. Configure Loopback 0:

```
interface Loopback0
ip address 1.1.1.1 255.255.255.255
```

3. Configure GigabitEthernet interfaces (repeat for each interface):

```
interface GigabitEthernet0/0
ip address 10.1.2.1 255.255.255.252
no shutdown
```

4. Save the configuration:

```
end
write memory
```

Repeat this process for all routers, using the appropriate IP addresses from the topology diagram.

Verify Basic Connectivity

After configuring all interfaces, verify basic connectivity:

1. From R1, ping the directly connected interface on R2:

```
R1# ping 10.1.2.2
```

2. Repeat similar tests between directly connected routers to ensure basic IP connectivity.

Save Your Progress

Once you've completed the initial setup and verified connectivity:

1. Save all router configurations: `write memory` on each router
2. Save your GNS3 project: File > Save project

Troubleshooting Tips

If you encounter issues during setup:

- Verify that all interfaces are in “up/up” state using `show ip interface brief`
- Check for typos in IP addresses and subnet masks
- Ensure that you've saved configurations on all devices
- Consult the GNS3 documentation if you have software-related issues

You are now ready to begin the OSPF configuration tasks. If you have any questions or issues, please consult with your instructor before proceeding to the next chapter.

Basic OSPF Configuration (Area 0 and Area 3)

Objectives

In this chapter, you will:

- Configure OSPF in Area 0 (backbone area)
- Set up OSPF in Area 3
- Implement OSPF authentication
- Configure OSPF network types
- Manipulate OSPF router IDs
- Control DR/BDR election

Configuring OSPF in Area 0

Follow these steps to configure OSPF in Area 0 on routers R1, R2, R3 and R4:

1. Enter global configuration mode:

```
configure terminal
```

2. Create the OSPF process with ID 1:

```
router ospf 1
```

3. Configure router ID (use loopback IP address):

```
router-id <loopback-ip>
```

4. Add networks to OSPF Area 0. For R1, this would look like:

```
network 1.1.1.1 0.0.0.0 area 0
network 10.1.2.0 0.0.0.3 area 0
network 10.1.3.0 0.0.0.3 area 0
network 10.1.33.0 0.0.0.3 area 0
```

5. Repeat steps 1-4 for R2, R3 and R4, using their respective IP addresses and networks.

Configuring OSPF in Area 3

Configure OSPF in Area 3 on router R5:

1. Enter global configuration mode on R5:

```
configure terminal
```

2. Create the OSPF process:

```
router ospf 1
```

3. Configure router ID:

```
router-id 5.5.5.5
```

4. Add networks to OSPF Area 3:

```
network 5.5.5.5 0.0.0.0 area 3
network 10.5.6.0 0.0.0.7 area 3
```

6. Repeat steps 1-4 for R1, R3 and R6, using their respective IP addresses and networks.

Implementing OSPF Authentication

Configure OSPF MD5 authentication on all interfaces in Area 0:

1. On each interface participating in OSPF Area 0, configure:

```
interface GigabitEthernet0/0
ip ospf message-digest-key 1 md5 ospfpass
ip ospf authentication message-digest
```

2. Enable authentication for the OSPF process:

```
router ospf 1
area 0 authentication message-digest
```

Repeat this for all routers in Area 0.

Configuring OSPF Network Types

Configure point-to-point OSPF network type on interfaces with /30 subnets:

1. On each applicable interface:

```
interface GigabitEthernet0/0
ip ospf network point-to-point
```

2. Repeat for all /30 interfaces in the topology.

Controlling DR/BDR Election

On the multi-access segment (10.5.6.0/29) connecting R1, R3, R5 and R6:

1. Configure R1 as DR:

```
interface GigabitEthernet2/0
ip ospf priority 200
```

2. Configure R3 as BDR:

```
interface GigabitEthernet2/0
```

```
ip ospf priority 100
```

3. Prevent R6 from participating in the election:

```
interface GigabitEthernet0/0
ip ospf priority 0
```

Verification and Troubleshooting

Use these commands to verify your configuration:

1. Check OSPF neighbours:

```
show ip ospf neighbor
```

2. Verify OSPF routes:

```
show ip route ospf
```

3. Examine OSPF database:

```
show ip ospf database
```

4. Check interface OSPF settings:

```
show ip ospf interface
```

Review Questions

1. Why is it important to manually set OSPF router IDs?
2. What is the purpose of OSPF authentication, and why is MD5 preferred over clear text?
3. In what scenarios would you choose to configure an OSPF network type as point-to-point?
4. Explain the roles of DR and BDR in OSPF, and why we might want to control their election.
5. What happens when a router with a higher priority than the DR or BDR is added?
6. On the segment S2 there are adjacencies with status 2WAY, explain the observed behaviour.
7. How does the point-to-point network type differ from the other OSPF network types?
8. Are hello packets for this network type (P2P) sent as unicast or multicast?
9. Does the P2P network type support the DR/BDR election? Why or why not?
10. Activate in R1 and R2 the `debug ip ospf events`. Change the hello timers on the interface between R1 and R2 to half of the current value. Wait for the dead interval. What was the impact from this change on the topology?
11. What is the relation between the hello and dead interval? Revert the hello timer to the previous value.

Lab Tasks

1. Complete all configurations as described above.
2. Verify full OSPF connectivity between all routers in Area 0 and Area 3.
3. Verify connectivity between PC22 and PC33.
4. Capture the output of all verification commands.
5. Answer all review questions.

Save your configurations and document your findings for your lab report.

OSPF Area 1 and Virtual Link Configuration

Objectives

In this chapter, you will:

- Configure OSPF Area 1
- Implement an OSPF virtual link
- Understand the purpose and limitations of virtual links
- Optimize route distribution
- Verify inter-area connectivity

Configuring OSPF in Area 1

Configure OSPF in Area 1 on routers R7 and R5:

1. On R7, enter global configuration mode and create the OSPF process:

```
configure terminal
router ospf 1
router-id 7.7.7.7
network 7.7.7.7 0.0.0.0 area 1
network 10.7.5.0 0.0.0.3 area 1
network 10.7.11.0 0.0.0.3 area 1
```

2. Similarly, on R5:

```
configure terminal
router ospf 1
network 10.7.5.0 0.0.0.3 area 1
```

Understanding the Need for a Virtual Link

In OSPF, all areas must be connected to the backbone (Area 0). However, in our topology, Area 1 is not directly connected to Area 0. To solve this, we'll create a virtual link through Area 3.

We'll create a virtual link between R5 (in Area 3) and R1 and R3 (in Area 0):

1. On R5:

```
router ospf 1
area 3 virtual-link 3.3.3.3 authentication message-digest
area 3 virtual-link 3.3.3.3 message-digest-key 1 md5 vlink
area 3 virtual-link 1.1.1.1 authentication message-digest
area 3 virtual-link 1.1.1.1 message-digest-key 1 md5 vlink
```

2. On R1 and R3:

```
router ospf 1
area 3 virtual-link 5.5.5.5 authentication message-digest
area 3 virtual-link 5.5.5.5 message-digest-key 1 md5 vlink
```

Note: The virtual link uses the router IDs, not the IP addresses of the interfaces. If you get errors on the virtual link establishment, check if the router-id's match, sometimes the errors point to unmatched area ID but they are on the router-id, a `clear ip ospf 1 process` will solve the problem.

Optimizing route distribution

Since area 1 sits on the end of the network, with a single connection to the network we can optimize the number of routes inside the network by converting Area 1 into a stub no-summary network.

1. On R5 and R7:

```
router ospf 1
area 1 stub no-summary
```

Check the different to the routing table with and without converting the area into an Stub no-summary area.

Verification and Troubleshooting

Use these commands to verify your configuration:

1. Check the status of the virtual link:

```
show ip ospf virtual-links
```

2. Verify OSPF neighbours:

```
show ip ospf neighbor
```

3. Check inter-area routes:

```
show ip route ospf
```

4. Examine the OSPF database:

```
show ip ospf database
```

Review Questions

1. What is the purpose of an OSPF virtual link?
2. Why is authentication important on a virtual link?
3. How does a virtual link affect the path that packets take through the network?
4. What are some limitations or potential issues with using virtual links?
5. What happens when area 0 becomes partitioned?
6. What does the IP address used in the virtual-link statement refer to?
7. Must this address be reachable before the virtual-link can be established?
8. How can we optimize the route tables of non-transit areas?

LAB TASKS

1. Complete all configurations as described above.
2. Verify full OSPF connectivity between all routers in Areas 0, 1, and 3.
3. Trace the path of a packet from R7 to R2. Document each hop.
4. Assess the ICMP connectivity between PC11 <-> PC22 and PC11 <-> PC33
5. Capture the output of all verification commands listed.
6. Answer all review questions.

Save your configurations and document your findings for your lab report.

OSPF AREA 2 AND EXTERNAL ROUTE REDISTRIBUTION

OBJECTIVES

In this chapter, you will:

- Configure OSPF Area 2
- Set up RIP on specified interfaces
- Redistribute RIP routes into OSPF
- Control LSA propagation
- Adjust OSPF metrics for external routes

CONFIGURING OSPF IN AREA 2

Configure OSPF in Area 2 on routers R2, R8 and R9:

1. On R9, enter global configuration mode and create the OSPF process:

```
configure terminal
router ospf 1
router-id 9.9.9.9
network 9.9.9.9 0.0.0.0 area 2
network 10.2.8.0 0.0.0.7 area 2
```

2. Similarly, on R8:

```
configure terminal
router ospf 1
router-id 8.8.8.8
network 8.8.8.8 0.0.0.0 area 2
network 10.2.8.0 0.0.0.7 area 2
```

3. Now, configure R2 to add Area 2.

CONFIGURING RIP

We'll configure RIP on R10 (and R9) to simulate external routes:

1. On R10 and R9:

```
router rip
version 2
network 10.0.0.0
no auto-summary
```

Redistributing RIP into OSPF and OSPF to RIP

Now, we'll redistribute the RIP routes into OSPF:

1. On R9:

```
router ospf 1
redistribute rip subnets
router rip
redistribute ospf 1 metric 1
```

Controlling LSA Propagation

We'll configure Area 2 as a Not-So-Stubby Area (NSSA) to control external LSA propagation:

1. On R8, R9 and R2:

```
router ospf 1
area 2 nssa
```

Adjusting OSPF Metrics for External Routes

Let's adjust the OSPF cost for redistributed routes:

1. On R9:

```
router ospf 1
redistribute rip subnets metric-type 1 metric 1000
```

Verification and Troubleshooting

Use these commands to verify your configuration:

1. Check OSPF neighbours:

```
show ip ospf neighbor
```

2. Verify OSPF and RIP routes:

```
show ip route
```

3. Examine the OSPF database, particularly Type-7 LSAs:

```
show ip ospf database nssa-external
```

4. Verify RIP configuration:

```
show ip protocols
```

Review Questions

1. What is the purpose of redistributing routes between routing protocols?
2. How does configuring an area as NSSA affect LSA propagation?
3. Why might we want to adjust metrics for redistributed routes?
4. What are the potential risks or challenges of route redistribution?
5. What different options are there for converting RIP external route costs on the OSPF?

Lab Tasks

1. Complete all configurations as described above.
2. Verify that the RIP routes are successfully redistributed into OSPF.
3. Trace the path of a packet from R1 to the S9 network. Document each hop.
4. Ensure that a ping from R10 with the loopback0 as a source interface will have connectivity to all the loopback routers on the topology. Ex: `R10# ping <other routers IP loopbacks> source loopback 0`.
5. Confirm connectivity between PC55 and all the other PCs.
6. Capture the output of all verification commands listed.
7. Answer all review questions.

Save your configurations and document your findings for your lab report.

Advanced OSPF Configurations and Maintenance Scenario

Objectives

In this chapter, you will:

- Optimize route table size through summarization
- Simulate a maintenance scenario
- Analyse OSPF behaviour during network changes

Route Summarization

Let's implement route summarization on R5 for routes it's advertising into OSPF:

1. On R5:

```
router ospf 1
area 1 range 10.2.0.0 255.255.0.0
```

This will summarize all routes from Area 2 into a single route when advertised into Area 0.

2. Check if summarization is working, explain why it’s not working.

Simulating a Maintenance Scenario

We'll simulate maintenance on the link between R1 and R3 by increasing its OSPF cost:

1. On R1, interface connected to R2:

```
interface GigabitEthernet0/0
ip ospf cost 1000
```

2. On R3, interface connected to R1:

```
interface GigabitEthernet0/0
ip ospf cost 1000
```

Verification and Analysis

Use these commands to verify your configuration and analyse the network behaviour:

1. Check OSPF routes before and after summarization:

```
show ip route ospf
```

2. Examine the OSPF database, focusing on summary LSAs:

```
show ip ospf database summary
```

3. Verify the path taken by traffic before and after the maintenance scenario:

```
trace <destination-ip>
```

4. Check OSPF interface costs:

```
show ip ospf interface
```

Review Questions

1. How does filtering summary LSAs affect the routing table and network visibility?
2. What are the benefits and potential drawbacks of route summarization?
3. How does changing OSPF interface costs impact routing decisions?
4. In what scenarios might you want to manipulate OSPF costs in a production network?

Lab Tasks

1. Complete all configurations as described above.
2. Compare the routing table on R9 before and after implementing the summary LSA filter. Document the differences.
3. Trace the path from R7 to R10 before and after the maintenance scenario. Document the path changes.
4. Capture the output of all verification commands listed, both before and after each configuration change.
5. Answer all review questions.
6. Write a brief analysis (1-2 paragraphs) on how the summarization and maintenance scenario affected network behaviour.

Save your configurations and document your findings for your lab report.

OSPF Optimization and Troubleshooting

Objectives

In this chapter, you will:

- Optimize OSPF convergence times
- Implement OSPF traffic engineering techniques
- Troubleshoot common OSPF issues
- Analyse OSPF behaviour under various failure scenarios

Optimizing OSPF Convergence

Let's implement some techniques to improve OSPF convergence times:

Enable OSPF fast hello packets on the link between R1 and R2:

1. On both R1 and R2 interfaces connecting them:

```
interface GigabitEthernet0/0
ip ospf dead-interval minimal hello-multiplier 4
```

OSPF Traffic Engineering

Implement OSPF traffic engineering to influence path selection:

1. Implement IP SLA tracking to dynamically adjust OSPF costs when a link gets congested:

On R3:

```
ip sla 1
 icmp-echo 10.3.4.[R4-g1/0-ipaddress] source-interface GigabitEthernet0/0
 frequency 5
 threshold 50
!
ip sla schedule 1 life forever start-time now
!
track 1 ip sla 1 state
!
interface GigabitEthernet0/0
 ip ospf cost 10
!
event manager applet CHANGE_OSPF_COST
 event track 1 state down
 action 1.0 cli command "enable"
 action 2.0 cli command "configure terminal"
 action 3.0 cli command "interface GigabitEthernet0/0"
 action 4.0 cli command "ip ospf cost 1000"
 action 5.0 cli command "end"
!
event manager applet RESTORE_OSPF_COST
 event track 1 state up
 action 1.0 cli command "enable"
```

```
action 2.0 cli command "configure terminal"
action 3.0 cli command "interface GigabitEthernet0/0"
action 4.0 cli command "ip ospf cost 10"
action 5.0 cli command "end"
```

Failure Scenario Analysis

Simulate and analyse these failure scenarios:

1. Link Congestion:
   - Do a trace from R7 to PC55 and at the same time do a ping between R3 and R4;
   - You should see on R3 messages stating that the track following "ip sla 1" is changing between up and down, if you don't see any changes check the current value with `show track 1` and adjust the ip sla 1 threshold to a more reasonable value;
   - Analyse the path changes on the trace command between ip sla coming up or down.
1. Link Failure:
   - Shut down the interface between R3 and R4;
   - Analyse the network's behaviour and convergence time.
2. Router Failure:
   - Simulate a failure of R3 by shutting it down
   - Observe and document the impact on the network

Verification and Analysis

Use these commands for verification and analysis:

1. Check OSPF neighbours and their states:

```
show ip ospf neighbor detail
```

2. Examine OSPF timers and LSA information:

```
show ip ospf
show ip ospf database
```

3. Verify OSPF interface settings:

```
show ip ospf interface brief
```

4. Analyse the IP routing table:

```
show ip route ospf
```

5. Use debugging for real-time OSPF analysis:

```
debug ip ospf events
debug ip ospf adj
```

(Remember to disable debugging when finished)

Review Questions

1. How do fast hello packets affect OSPF convergence?
2. What are the potential benefits and risks of OSPF traffic engineering?
3. Describe the process you would follow to troubleshoot an OSPF neighbour relationship issue.
4. How can you use IP SLA with OSPF to create more resilient networks?

Lab Tasks

1. Implement all optimizations and traffic engineering configurations as described.
2. Complete each troubleshooting scenario, documenting your process and solutions.
3. Analyse and document the network's behaviour during each failure scenario.
4. Capture relevant output from the verification commands for each major configuration change and scenario.
5. Answer all review questions.
6. Write a brief report (2-3 paragraphs) discussing the impact of the optimizations and your observations from the troubleshooting and failure scenarios.

Save your configurations and include all documentation in your lab report.

Conclusion

Congratulations on completing the Advanced OSPF Configuration Lab! Throughout this lab, you've gained hands-on experience with:

- Configuring multi-area OSPF networks
- Implementing OSPF authentication
- Setting up virtual links
- Controlling route propagation and summarization
- Redistributing routes between OSPF and other protocols
- Optimizing OSPF performance and convergence
- Troubleshooting common OSPF issues

These skills form the foundation of designing, implementing, and maintaining complex OSPF networks in real-world scenarios.

Appendices

OSPF command reference

Basic OSPF Configuration

1. Enable OSPF process:

```
router ospf <process-id>
```

2. Configure Router ID:

```
router-id <A.B.C.D>
```

3. Enable OSPF on an interface:

```
network <ip-address> <wildcard-mask> area <area-id>
```

4. Manually specify OSPF on an interface:

```
ip ospf <process-id> area <area-id>
```

OSPF Area Configuration

1. Configure stub area:

```
area <area-id> stub
```

2. Configure totally stubby area:

```
area <area-id> stub no-summary
```

3. Configure Not-So-Stubby Area (NSSA):

```
area <area-id> nssa
```

4. Configure NSSA with no summary:

```
area <area-id> nssa no-summary
```

OSPF Authentication

1. Enable MD5 authentication on an interface:

```
ip ospf message-digest-key <key-id> md5 <key>
ip ospf authentication message-digest
```

2. Enable area-wide authentication:

```
area <area-id> authentication message-digest
```

OSPF Timers and Optimization

1. Modify Hello and Dead intervals:

```
ip ospf hello-interval <seconds>
ip ospf dead-interval <seconds>
```

2. Configure OSPF fast hello packets:

```
ip ospf dead-interval minimal hello-multiplier <multiplier>
```

3. Adjust LSA and SPF throttle timers:

```
timers throttle lsa all <start-interval> <hold-interval> <max-interval>
```

```
timers throttle spf <start-interval> <hold-interval> <max-interval>
```

OSPF Route Summarization and Filtering

1. Summarize routes at an ABR:

```
area <area-id> range <address> <mask>
```

2. Summarize external routes:

```
summary-address <address> <mask>
```

3. Filter routes using prefix list:

```
area <area-id> filter-list prefix <prefix-list-name> in/out
```

OSPF Virtual Links

1. Configure a virtual link:

```
area <transit-area-id> virtual-link <router-id>
```

OSPF Route Redistribution

1. Redistribute routes into OSPF:

```
redistribute <protocol> subnets
```

2. Set default metric for redistributed routes:

```
default-metric <metric>
```

OSPF Network Types

1. Set OSPF network type:

```
 ip ospf network {broadcast | non-broadcast | point-to-multipoint | point-to-point}
```

OSPF Cost and Priority

1. Set OSPF cost on an interface:

```
ip ospf cost <cost>
```

2. Set OSPF priority for DR/BDR election:

```
ip ospf priority <priority>
```

OSPF Verification and Troubleshooting

1. Display OSPF neighbours:

```
show ip ospf neighbor
```

2. Show OSPF database:

```
show ip ospf database
```

3. Display OSPF interface information:

```
show ip ospf interface
```

4. Show OSPF routes:

```
show ip route ospf
```

5. Debug OSPF operations:

```
debug ip ospf events
debug ip ospf adj
```

Remember to use these commands responsibly, especially debugging commands in production environments. Always verify the impact of configuration changes before applying them to live networks.

TROUBLESHOOTING GUIDE

COMMON OSPF ISSUES AND SOLUTIONS

OSPF Neighbor Adjacency Problems

Symptoms:

- Neighbours stuck in INIT or 2-WAY state
- "OSPF-5-ADJCHG" messages in logs

Possible Causes and Solutions:

a) Mismatched Area Numbers

- Verify area configuration: `show ip ospf interface`
- Correct area assignments if necessary

b) Authentication Mismatch

- Check authentication settings: `show ip ospf interface`
- Ensure consistent authentication across neighbors

c) Mismatched Hello/Dead Timers

- Verify timers: `show ip ospf interface`
- Align timers on both sides of the link

d) MTU Mismatch

- Check interface MTU: `show interface`
- Adjust MTU to match on both sides

e) Misconfigured Network Type

- Verify network type: `show ip ospf interface`
- Ensure consistent network types on the link

Missing OSPF Routes

Symptoms:

- Expected routes not in routing table
- Incomplete connectivity

Possible Causes and Solutions:

a) Network Statement Missing

- Review OSPF configuration: `show run | section router ospf`
- Add missing network statements

b) Passive Interface

- Check for passive interfaces: `show ip ospf interface`
- Remove passive setting if unintended

c) Route Filtering

- Examine any distribute-lists or prefix-lists
- Adjust filters if blocking desired routes

d) Area Type Restrictions

- Verify area type (stub, NSSA): `show ip ospf`
- Ensure area type allows necessary route types

Suboptimal Routing

Symptoms:

- Traffic taking unexpected paths
- Increased latency or congestion

Possible Causes and Solutions:

a) Incorrect OSPF Costs

- Review interface costs: `show ip ospf interface`
- Adjust costs to influence route selection

b) Unintended Equal-Cost Multi-Path (ECMP)

- Check for multiple equal-cost paths: `show ip route`

- Adjust costs if single path preferred

c) Route Summarization Issues

- Examine summary addresses: `show ip ospf`
- Refine summarization to avoid suboptimal paths

OSPF Process Issues

Symptoms:

- OSPF process not running
- Unexpected CPU utilization

Possible Causes and Solutions:

a) OSPF Process Crash

- Check OSPF process status: `show ip protocols`
- Review logs for crash information
- Consider enabling OSPF graceful restart

b) High CPU Utilization

- Monitor CPU usage: `show processes cpu`
- Optimize network design, consider summarization
- Tune SPF and LSA timers if necessary

External Route Issues

Symptoms:

- External routes not propagating
- Unexpected external route behaviour

Possible Causes and Solutions:

a) Redistribution Problems

- Verify redistribution configuration: `show run | section redistribute`
- Check ASBR status: `show ip ospf database asbr-summary`

b) NSSA Configuration

- Ensure proper NSSA setup: `show ip ospf`
- Verify Type-7 to Type-5 LSA conversion

General Troubleshooting Steps

1. Verify Physical Connectivity
   - Check interface status: `show ip interface brief`

- Ensure cables are properly connected

2. Review OSPF Configuration
   - Examine OSPF process settings: `show run | section router ospf`
   - Verify interface OSPF settings: `show ip ospf interface`
3. Check OSPF Neighbour Status
   - View neighbour adjacencies: `show ip ospf neighbor`
   - Look for any neighbours stuck in INIT or 2-WAY state
4. Examine OSPF Database
   - Review LSA information: `show ip ospf database`
   - Ensure expected LSAs are present and correct
5. Analyse IP Routing Table
   - Check for OSPF routes: `show ip route ospf`
   - Verify next-hop information and metrics
6. Use Debug Commands
   - Enable OSPF debugging: `debug ip ospf events`, `debug ip ospf adj`
   - Monitor real-time OSPF behaviour
7. Review System Logs
   - Check for OSPF-related messages: `show logging`

Remember: Always approach troubleshooting systematically, making one change at a time and verifying the impact before moving to the next step. Document all changes for easy rollback if needed.

Lab Report Guidelines and Grading

Suggested Outline and Content of the Lab Report

1. Title Page

- Lab title.
- Group member names and student IDs.
- Group number.
- Date of submission.

2. Table of Contents

3. Introduction (5% of report)

- Brief overview of the lab objectives.
- Your understanding of OSPF and its importance in networking.

4. Methodology (10% of report)

- Description of the lab environment (software used, network topology, IP addressing).
- Brief explanation of the approach taken for each major task.

5. Results and Analysis (50% of report)

For each major section of the lab:

- Configuration snippets (key commands used).

- Screenshots of critical outputs (e.g., routing tables, OSPF databases).
- Analysis of the results (What do they mean? How do they relate to OSPF concepts?).
- Any challenges encountered and how they were resolved.

6. Advanced OSPF Concepts (15% of report)

- Detailed explanation of advanced OSPF concepts encountered in the lab.
- Real-world applications of these concepts.

7. Troubleshooting Section (10% of report)

- Description of any issues encountered during the lab.
- Step-by-step explanation of your troubleshooting process.
- Resolution of the issues.

8. Conclusion (5% of report)

- Summary of key learnings.
- Reflections on the lab experience.
- Suggestions for future improvements to the lab.

9. References

- Any external sources consulted during the lab or report writing.

10. Appendices

- Full configuration files for all routers and VPCs.
- Any additional relevant outputs or diagrams.

Formatting

- Use a clear, readable font (e.g., Arial, Calibri, Times New Roman).
- Font size: 11 or 12 point.
- 1.5 line spacing.
- Page numbers on all pages except the title page.
- Clear headings and subheadings.
- Proper labelling of all figures and tables.

Grading Criteria

1. Technical Accuracy (40%)

- Correct OSPF configurations.
- Accurate interpretation of routing tables and OSPF databases.
- Proper implementation of advanced OSPF features.

2. Analysis and Understanding (25%)

- Depth of analysis in the Results section.
- Demonstration of understanding OSPF concepts.
- Quality of insights in the Advanced OSPF Concepts section.

3. *Troubleshooting Skills (15%)*

- Identification of issues.
- Logical approach to problem-solving.
- Effectiveness of solutions.

4. *Report Structure and Clarity (10%)*

- Clear organization of the report.
- Logical flow of ideas.
- Proper use of technical language.

5. *Documentation and Presentation (10%)*

- Inclusion of all required sections.
- Quality of diagrams and screenshots.
- Adherence to formatting requirements.

6. *Bonus Points (up to 5% extra)*

- Exceptional insights or analysis.
- Creative solutions to complex problems.
- Going beyond the basic requirements of the lab.

Note: Plagiarism or copying configurations from other students will result in a zero grade for the lab report. All work must be your own.

Documentation

http://www.cisco.com/c/en/us/support/docs/ip/open-shortest-path-first-ospf/13692-21.html

https://supportforums.cisco.com/document/116011/ospf-inter-area-route-summarization

http://www.cisco.com/c/en/us/support/docs/ip/open-shortest-path-first-ospf/47870-ospfdb11.html

http://www.cisco.com/c/en/us/support/docs/ip/open-shortest-path-first-ospf/16437-ospfdbtoc.html?referring_site=bodynavhttps://rednectar.net/archives/vpcs-tutorial